

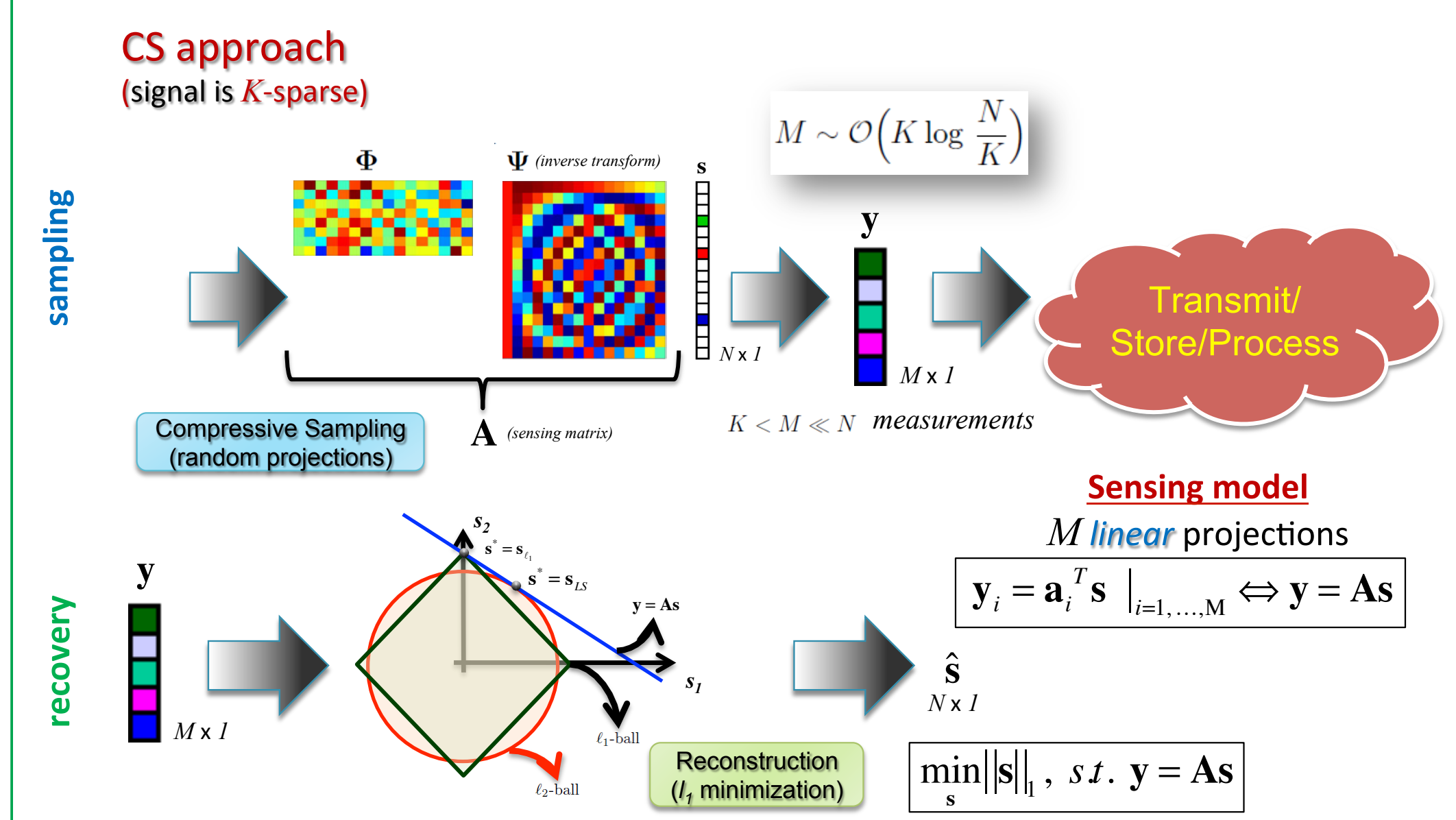
OBJECTIVE

- Joint wideband localization and acquisition of acoustic sources
 - Assume a sparse recovery problem: discretize the acoustic scene into a grid of predefined dimensions
 - Handles off-grid source locations: accounts for energy leakage effect

CONTRIBUTIONS

- Assume sources are located away from the grid points
- Cannot arbitrarily increase the grid size: Restricted Isometry Property (RIP) violation
- Generate a grid which is shifted across the entire acoustic scene (grid-shift process)
- Adopt Orthogonal Matching Pursuit (OMP)-based sparse recovery: real-time framework

COMPRESSIVE SENSING (CS)



- Recovery guarantee with high probability if RIP of order K is satisfied for small values of δ_K

$$(1 - \delta_K) \|s\|_2^2 \leq \|As\|_2^2 \leq (1 + \delta_K) \|s\|_2^2$$

- Tackle computational intractability of RIP by using the coherence of A

$$\mu = \max_{1 \leq i, j \leq N} \frac{|a_i^H a_j|}{\|a_i\|_2 \|a_j\|_2},$$

CS KEY POINTS

- The lower the coherence the higher the probability of an accurate sparse vector estimation
 - Linear independence among the columns of A
- Increased size of vector $s \Rightarrow$ higher number of columns in $A \Rightarrow$ larger inter-column linear dependence $\Rightarrow$ RIP/coherence violation $\Rightarrow$ decreased sparse recovery performance
- Basis mismatch between the continuous physical model and the discretized assumed model reflected in $A \Rightarrow$ introduce the concept of grid-shift

PROPOSED METHOD

- Location estimation task formulated as a sparse recovery problem
- Time-frequency representation of the received signal at the m -th microphone

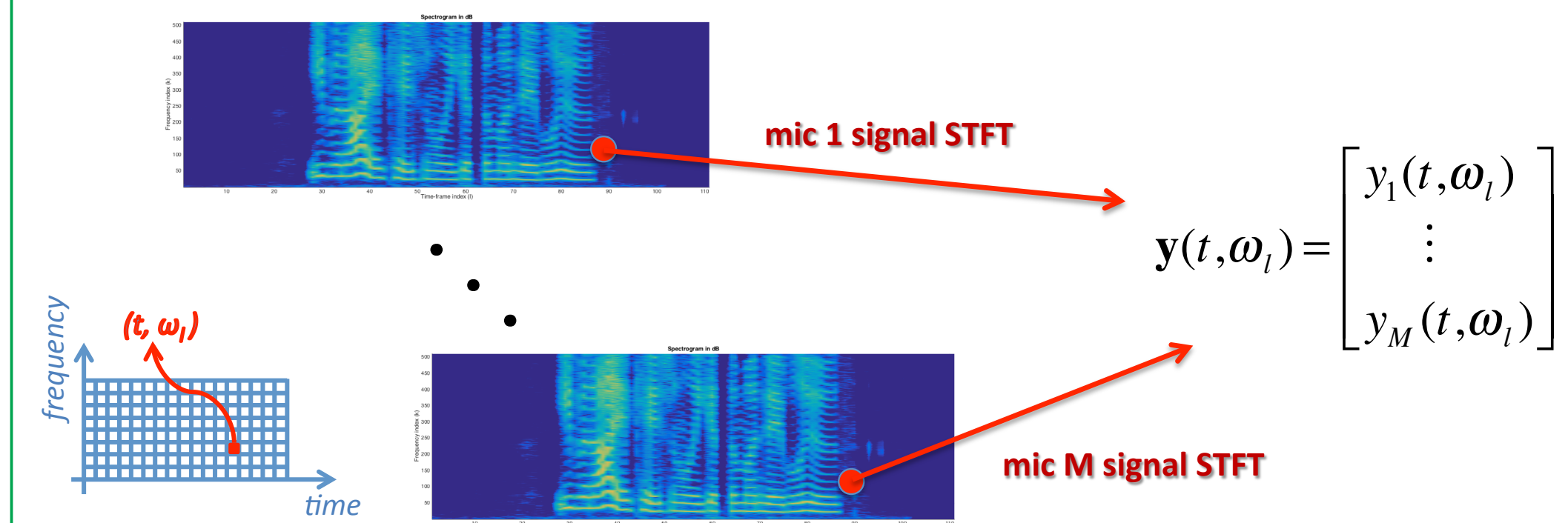
$$y_m(t, \omega_l) = \sum_{i=1}^K A_{m,i}(\omega_l) x_i(t, \omega_l) + n_m(t, \omega_l)$$

- M : number of microphones (randomly placed in the acoustic scene)
- K : number of acoustic sources ($K < M$)
- $x_i(t, \omega_l)$: i -th source signal
- $n_m(t, \omega_l)$: noise at microphone m
- $\omega_l = 2\pi f$: angular frequency (with f_l denoting the frequency in Hz)
- $A_{m,i}(\omega_l)$: acoustic transfer function between microphone m and source i
- Assume a discretization of the area of interest into $N \gg K$ grid points

- Introduce an activation vector of the form

$$s_n(t, \omega_l) = \begin{cases} x_i(t, \omega_l), & \text{if source } i \text{ is active at grid point } n \\ 0, & \text{otherwise} \end{cases}$$

- $s(t, \omega_l)$ is a K -sparse vector



- Using matrix form notation the observation vector can be written as

$$\mathbf{y}(t, \omega_l) = \mathbf{A}(\omega_l) \mathbf{s}(t, \omega_l) + \mathbf{n}(t, \omega_l)$$

- $\mathbf{y}(t, \omega_l) \in \mathbb{C}^M$: observations vector
- $\mathbf{s}(t, \omega_l) \in \mathbb{C}^N$: unknown vector of the complex source amplitudes at all N grid points
- $\mathbf{n}(t, \omega_l) \in \mathbb{C}^M$: additive noise error term
- $\mathbf{A}(\omega_l) \in \mathbb{C}^{M \times N}$: acts as a mapping between $\mathbf{s}(t, \omega_l)$ and $\mathbf{y}(t, \omega_l)$
- $K < M \ll N \Rightarrow$ few sources generate the acoustic field $\Rightarrow$ undetermined problem
- Recovery of the sparse activation vector $\mathbf{s}(t, \omega_l)$ given $\mathbf{y}(t, \omega_l)$ and $\mathbf{A}(\omega_l)$ via ℓ_1 -norm minimization problem

$$\hat{\mathbf{s}}(t, \omega_l) = \arg \min_{\mathbf{s}} \|\mathbf{s}\|_1, \text{ s.t. } \|\mathbf{y}(t, \omega_l) - \mathbf{A}(\omega_l) \mathbf{s}\|_2 \leq \varepsilon$$
 - ε : noise level
 - Infer each source location by the index of non-zero elements in $\hat{\mathbf{s}}(t, \omega_l)$

ACOUSTIC SOURCE LOCALIZATION

- Introduce the concept of grid-shift to balance between RIP violation and energy leakage phenomena
- $\mathbf{A}_r(\omega_l) = [\mathbf{a}(\mathbf{p}_1^r) | \mathbf{a}(\mathbf{p}_2^r) | \dots | \mathbf{a}(\mathbf{p}_N^r)] \in \mathbb{C}^{M \times N}$ sensing matrix corresponding to the r -th grid $\mathbf{G}_r = [\mathbf{p}_1^r | \mathbf{p}_2^r | \dots | \mathbf{p}_N^r]$
 - $\mathbf{p}_n^r \in \mathbb{R}^3$ (with $n = 1, \dots, N$) are the coordinates of the grid points corresponding to the r -th grid

Input: $\mathbf{y}(t, \omega_l)$, $t = 1, \dots, T$, $l = 1, \dots, F$

grids $\mathbf{G}_r$, $r = 1, \dots, R$, where R is the total number of grid-shifts, tolerance ε , number of sources K

Output: estimated source locations

- for $r = 1 : R$ //loop over grid shifts
- for $t = 1 : T$ //loop over time frames
- for $f = 1 : F$ //loop over frequency bins
- build matrix $\mathbf{A}_r(\omega_l)$ using the current grid $\mathbf{G}_r$
- $\hat{\mathbf{s}}(t, \omega_l) = \arg \min_{\mathbf{s}} \|\mathbf{s}\|_1, \text{ s.t. } \|\mathbf{y}(t, \omega_l) - \mathbf{A}_r(\omega_l) \mathbf{s}\|_2 \leq \varepsilon$
- $\mathbf{L}_r(t, \omega_l, :) \leftarrow \hat{\mathbf{s}}(t, \omega_l)^T$, where $\mathbf{L}_r \in \mathbb{C}^{T \times F \times N}$
- end
- end
- compute the magnitude of sparse solutions $|\mathbf{L}_r|$
- compute the mean $\boldsymbol{\mu}_r \in \mathbb{R}^{N \times 1}$ corresponding to all the $T \times F$ sparse solutions $|\mathbf{L}_r|$
- find the indices $\{\hat{i}_1^r, \dots, \hat{i}_K^r\}$ of the top- K (maximum) values of the mean $\boldsymbol{\mu}_r$
- find the current source position estimates $\hat{\mathbf{Q}}_r = [\mathbf{G}_r(:, \hat{i}_1^r), \dots, \mathbf{G}_r(:, \hat{i}_K^r)] \in \mathbb{R}^{3 \times K}$
- end
- stack all the estimated source positions into one matrix $\hat{\mathbf{Q}} = [\hat{\mathbf{Q}}_1^T \dots \hat{\mathbf{Q}}_R^T]^T \in \mathbb{R}^{3 \times RK}$
- find source locations $\hat{\mathbf{P}} \in \mathbb{R}^{3 \times K}$ by K -means

ACOUSTIC SOURCE ACQUISITION

- Sparse opt. problem solved for all the time-frequency bins focusing on the estimated source positions

Input: estimated source locations $\hat{\mathbf{P}}$, grid $\mathbf{G}$ of size N centered in the middle of the acoustic scene, tolerance ε , number of sources K

Output: separated sources (in the time domain)

- find the indices $g_1, \dots, g_K$ of the grid points in $\mathbf{G}$ which are closest to the estimated source locations $\hat{\mathbf{P}}$
- obtain a new grid $\tilde{\mathbf{G}}$ (from grid $\mathbf{G}$) by replacing the K grid points found in the previous step with the source location estimates $\hat{\mathbf{P}}$
- for $t = 1 : T$ //loop over time frames
- for $f = 1 : F$ //loop over frequency bins
- build matrix $\tilde{\mathbf{A}}(\omega_l)$ using the grid $\tilde{\mathbf{G}}$
- $\hat{\mathbf{s}}(t, \omega_l) = \arg \min_{\mathbf{s}} \|\mathbf{s}\|_1, \text{ s.t. } \|\mathbf{y}(t, \omega_l) - \tilde{\mathbf{A}}(\omega_l) \mathbf{s}\|_2 \leq \varepsilon$
- $\mathbf{L}(t, \omega_l, :) \leftarrow \hat{\mathbf{s}}(t, \omega_l)^T$
- end
- end
- for $k = 1 : K$ //loop over each source
- compute the inverse short-time Fourier transform (STFT) of $\mathbf{L}(:, :, g_k)$ to obtain the k -th source time-domain signal
- end

EXPERIMENTAL RESULTS

- Image-Source method [1] $\Rightarrow$ box-shaped room of dimensions $6 \times 3.5 \times 3$ meters
- 10 microphones placed uniformly at random using all the space of dimensions $1.2 \times 0.7 \times 0.6$ meters located in the center of the room
- speech acoustic sources (randomly placed) of 3 seconds sampled at 8 kHz: randomly selected from the VOICES corpus [2]
 - frame length=640 samples, overlap=50%, FFT size=1024
- White Gaussian noise source signals
- Grid size: $7 \times 5 \times 6$
- Step size between each pair of neighboring grid points: $\Delta \mathbf{p} = [0.75, 0.58, 0.42]$ meters along x-axis, y-axis and z-axis
- Total number of grid-shifts: $R = 8$
- Proposed approach vs. perturbed-OMP [3]
- Proposed approach vs. MVDR-based beamformer [4]

Table 1: RMSE localization errors (in meters) for the grid-shift (GS) and perturbed-OMP (POMP) in the case of an anechoic environment.

SNR (dB)	two sources		three sources		four sources	
	GS	POMP	GS	POMP	GS	POMP
5	0.2727	0.3864	0.4804	0.5519	0.3526	0.6528
10	0.3526	0.5626	0.6642	0.6067	0.4023	0.7309
15	0.3017	0.3908	0.3742	0.4963	0.4440	0.5609
20	0.3517	0.2828	0.3675	0.3775	0.3797	0.4385
25	0.3244	0.2464	0.3414	0.3744	0.3560	0.6497

Table 2: RMSE angle-of-arrival errors (in degrees) for the grid-shift (GS) and MVDR in the case of a reverberant environment.

SNR (dB)	two sources		three sources		four sources	
	GS	MVDR	GS	MVDR	GS	MVDR
5	3.5230	4.8109	4.3875	6.5245	5.1718	6.1696
10	3.7004	4.7953	3.8504	5.1604	5.7962	6.1664
15	3.8226	5.7409	4.2013	5.8668	4.9736	7.4858
20	3.7263	5.0382	5.0752	6.5678	4.7431	7.1989
25	3.2619	4.8438	4.4231	6.5934	4.4616	7.4194

- BSSEval toolbox [5]: Signal-to-Distortion Ratio (SDR), Signal-to-Interference Ratio (SIR), Signal-to-Artifacts Ratio (SAR)

Table 3: Separation performance: no noise, no reverberation.

metric (dB)	two sources	three sources	four sources
SDR	5.7017	2.9224	2.1949
SIR	18.6001	13.7273	11.0495
SAR	6.3163	4.3916	3.9460

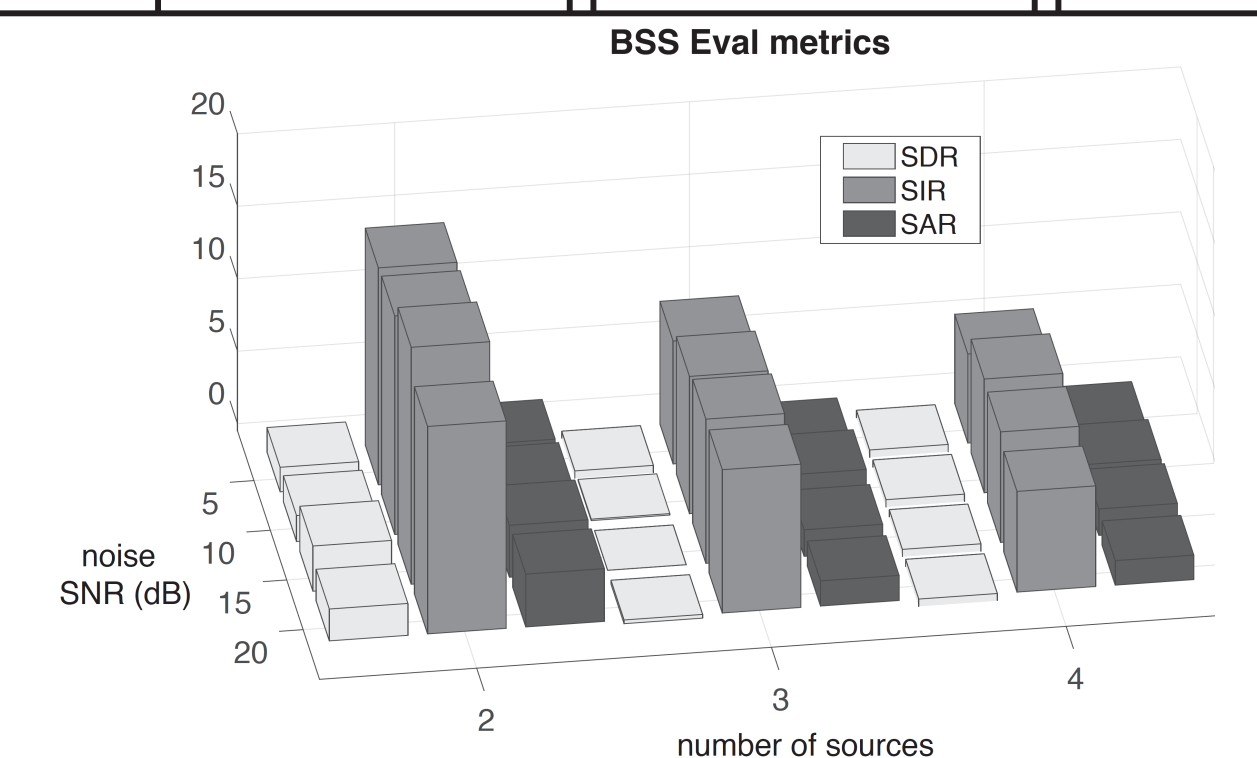


Figure 1: Separation performance: added white noise (SNR 5, 10, 15, 20 dB), reverberation time 259 msec.

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